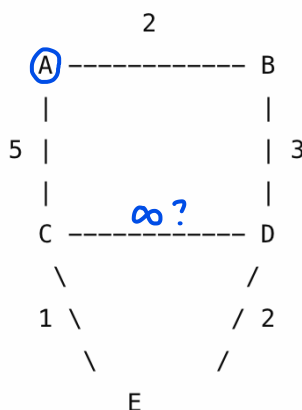


Exercise 2: Link State and Distance Vector Routing

Name: Norraphat Wattanachote Student ID: 6710405001 Date: 21/9/26

Learning objectives: Apply Dijkstra and Bellman-Ford; construct forwarding tables; analyze convergence, failures, and count-to-infinity.

Network topology:



Part 1 — Link State Routing

Question 1 — Dijkstra's Algorithm

Starting at router A, execute Dijkstra's algorithm. Show every update and the node confirmed at each step.

Step	Confirmed Node	D(B), Prev.	D(C), Prev.	D(D), Prev.	D(E), Prev.
0	A	∞	∞	∞	∞
1	B	2 A	∞	∞	∞
2	C	2 A	5 A	∞	∞
3	D	2 A	5 A	5 B	∞
4	E	2 A	5 A	5 B	6 C

Question 2 — Shortest-Path Tree

Draw the shortest-path tree rooted at A, then list the shortest path and total cost.

SPT sketch:

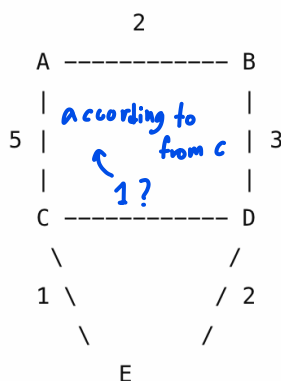
Destination	Shortest Path	Total Cost
B	A → B	2
C	A → C	5
D	A → B → D	5
E	A → C → E	6

Question 3 — Forwarding Table

Destination	Shortest Cost	Next Hop	Complete Path
B	2	B	A→B
C	5	C	A→C
D	5	B	A→B→D
E	6	C	A→C→E

Part 2 — Distance Vector Routing

Consider the same network. Initially, each router knows only the cost to its directly connected neighbors.



Question 4 — Bellman-Ford Update

Router A receives the following distance vectors from its neighbors:

From B:

A=2, B=0, C=4, D=3, E=5;

From C:

A=5, B=4, C=0, D=1, E=1.

Use the Bellman-Ford equation. Calculate router A's new distance to:

Show all calculations, including the neighbor through which the minimum-cost route is obtained.

Destination	Via B calculation	Via C calculation	Minimum Cost	Next Hop
B	$2 + 0 = 2$	$5 + 4 = 9$	2	B
C	$2 + 4 = 6$	$5 + 0 = 5$	5	C
D	$2 + 3 = 5$	$5 + 1 = 6$	5	B
E	$2 + 5 = 7$	$5 + 1 = 6$	6	C

Question 5 — Distance Vector Table

After receiving the distance vectors, construct router A's updated routing table.

Destination	Next Hop	Distance	Full Path
B	B	2	A→B
C	C	5	A→C
D	B	5	A→B→D
E	C	6	A→C→E

Part 3 — Routing Convergence

Question 6 — Link Failure

The C-D link fails. Explain separately what happens with Link State and Distance Vector routing.

Link State routing.

Your answer should describe:

- Which router(s) detect the failure.
- What information is flooded through the network.
- How routers update their link-state databases.
- How routers calculate new shortest paths.

Distance Vector routing.

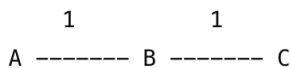
Your answer should describe:

- How routers learn about the failure.
- How updated distance vectors propagate.
- Why routers may temporarily have incorrect routing information.
- How routing eventually converges.

Link State Diststry	Distance Vector Bell-Ford
C, D ได้รับ Hello packet มาจาก V	C D พบ V มาเจอเพื่อนแล้ว
ส่ง LSA ไปให้ flood ทุก router ใน network	C D ไป V มาเจอเพื่อนแล้วบอกเพื่อน
ทุก router update link state ที่ตัวเอง	router ส่งไป V มาเจอเพื่อน
ทุก router มี dist table ใหม่	ตารางที่เก็บไว้กับเพื่อนบอก route ที่มันไปเจอเพื่อน loop
Converg ใช้เวลานานกว่าคือช้า	จน converg ถึงพร้อมคือใน loop update ที่ตัวเอง

Part 4 — Count-to-Infinity

Consider



Initially A reaches C through B at cost 2. The B–C link fails, but B receives stale information from A saying C is reachable at cost 2.

Question 7

- Why can this create a routing loop? $A \xrightarrow{1} B \Rightarrow B \xrightarrow{101} A \Rightarrow A \xrightarrow{10101} B \Rightarrow B \xrightarrow{1010101} A$ *101010101... B ကဲ့သို့ A သို့မဟုတ် A သို့မဟုတ် B သို့မဟုတ် C ကဲ့သို့*
- Show how the estimated distance to C can increase over successive iterations.
- What is count-to-infinity? $\rightarrow$ *ကိန်းဂဏန်းများသည် အဆင့်မြင့်မီသော အဆင့်သို့ ရောက်ရှိရန် မဖြစ်နိုင်ပါ။*
- Why does it occur in Distance Vector routing? *101010101... သို့မဟုတ် local လိုက်နာမှုကို echo chamber ကဲ့သို့ ခြေထောက်ပေးသည်။*
- Name a mitigation approach.

Part 5 — Compare LS and DV

Question 8

Characteristic	Link State	Distance Vector
Basic algorithm	find shortest path through all other nodes	find shortest path through neighbor node
Information maintained	Link state DB	Distance vector
Information exchanged	full network topology	distance to each neighbor V via neighbor
Algorithm	Dijkstra	Bellman-ford
Topology knowledge	global	local
Routing updates	flood to all routers	update only when hop
Convergence	fast	slow
Routing loops	X	✓
Count-to-infinity	X	✓
Example protocol	IS-IS	BGP

Part 6 — Design Question

Question 9

You are designing a routing system for an enterprise with 500 routers requiring fast convergence, accurate topology knowledge, efficient routing, and hierarchical design. Choose LS or DV and justify your choice with at least three technical arguments. Your answer should discuss at least three advantages and/or disadvantages of your chosen approach.

if LS is chosen - converge is / network is global in accurate topology / route efficient /

disadvantage - if memory in router is / in 1 router = flood of updates

Challenge — Algorithmic Analysis

Question 10

For N routers and E links, compare LS and DV in ¹information maintained, ²communication overhead, ³computation, ⁴convergence, and ⁵scalability. Explain why large-scale intra-domain routing generally favors Link State.

1	LS: $O(N+E)$	DV: $O(N \cdot \text{degree})$	in LS it's more efficient
2	LS: $O(N \cdot E)$ full network flood	DV: $O(N \cdot E)$ slowly update	LS scale better than DV in large area network
3	LS: $O(E \log N)$	DV: $O(N \cdot \text{degree})$	LS is bandwidth efficient DV is not route computation efficient
4	LS: $O(1)$	DV: $O(N)$	LS is faster loop free routing
5	LS: hierarchical area design	DV: hop count limitation	